

Non-distortionary belief elicitation

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- Many experiments involve incentivized truthful reporting of beliefs
 - examples: self-confidence, cognitive uncertainty, information acquisition, dynamic games,
- Incentives improve accuracy (Schlag et al. (2015) and others).
- Often, questions are *action-dependent*:
 - expected payoffs, expected regret, likelihood of being correct.
- But incentivization of *action-dependent questions* may distort behavior.

Example

A subject writes an IQ test with $N \geq 2$ True-False problems:

$$u(a, \theta) = \sum_{i \leq N} 1\{a_i = \theta_i\}.$$

Questions about expected value of $X(a, \theta)$:

- *What is your expected score?* $\implies X(a, \theta) = u(a, \theta)$.
It can be incentivized using Becker-DeGroot-Marschak (BDM) scoring rule.
- *What is the likelihood your score is above 50%?*
 $\implies X(a, \theta) = 1\{u(a, \theta) \geq \frac{N}{2}\}$
Without restrictive assumptions on beliefs, **any incentivization will distort test answers.**

- Distortions are problematic:
 - they affect interpretation of experimental results,
 - make honest and quiet instructions (Danz et al. (2022)) difficult ,
 - ethical and legal issues,
 - past literature: "moral hazard" in (Chambers and Lambert, 2021), "hedging" (Blanco et al. (2010)), "contamination" (Healy (2024)).
- Typical solutions:
 - don't tell the subject.
 - don't incentivize.
 - assume narrow-bracketing.
 - ask (many) *action-independent* questions Chen et al. (2026).

Our solution: choose the question better!

Research objective

How to incentivize belief elicitation without distortion? When can it be done?

Answer

- Sufficient conditions: questions about expected payoffs or "affine" transformations thereof (e.g. expected regret) can be incentivized.
- A complete characterization in three classes of decision problems.
- Methods: mechanism design, linear algebra, and graphs.

- Bayesian decision problem:

$$\max_a \mathbb{E}_p u(a, \theta)$$

- $u(a, \theta)$ - payoffs
- $p \in \Delta\Theta$ - privately known belief,
- static problem, no learning, actions do not affect states,
- DM is asked to report $r = \mathbb{E}_p X(a, \cdot)$ (one question only!),

Example

- expected payoffs: $X(a, \theta) = u(a, \theta)$
- expected regret: $X(a, \theta) = \max_b u(b, \theta) - u(a, \theta)$
- (ex post) correct choice: $X(a, \theta) = \begin{cases} 1 & \text{if } a \in \arg \max_{b \in A} u(b; \theta) \\ 0 & \text{otherwise.} \end{cases}$
- probability of state θ_0 : $X(a, \theta) = \mathbb{1}\{\theta = \theta_0\}$
- $Y(a, \theta) = \gamma(a)X(a, \theta) + \kappa(a)$ is *equivalent* to X for $\gamma(a) \neq 0$.

- Experimenter designs a scoring rule $V(a, r, \theta)$ so that:

$$\max_{a, r} \mathbb{E}_p V(a, r, \theta),$$

- benchmark case: experimentalist knows payoffs $u(\cdot)$ and observes a and θ ,
- V , typically, will be related to u (but it does not have to be!).

Incentivizability

Question X is *incentivizable* if there exists a scoring rule V such that

$$\arg \max_{a, r} \mathbb{E}_p V(a, r, \cdot) = \left\{ (a, \mathbb{E}_p X(a; \cdot)) : a \in \arg \max_{b \in A} \mathbb{E}_p u(b; \cdot) \right\},$$

Sufficient conditions

Lemma

The following questions (and their equivalents) are incentivizable: for any $d \in \mathbb{R}^\Theta$

- $X(a, \theta) = d(\theta)$,
- $X(a, \theta) = u(a, \theta) + d(\theta)$.

Becker-DeGroot-Marschak (1964):

- W.l.o.g. normalize $0 < X(a, \theta) < 1$.
- subject reports $r \in [0, 1]$,
- random number $x \in [0, 1]$ is drawn uniformly,
- if $x \leq r$, the subject receives $X(a, \theta)$,
- otherwise, if $r \leq x$, the subject receives x .

$X(a, \theta)$	☒	☐	0
$X(a, \theta)$	☒	☐	0.01
...			...
$X(a, \theta)$	☒	☐	r
$X(a, \theta)$	☐	☒	$r + 0.01$
...			...
$X(a, \theta)$	☐	☒	1

$$\max_r \mathbb{E}_p V(r, a, \cdot) = \frac{1}{2}(\mathbb{E}_p X)^2 + \frac{1}{2} = \frac{1}{2}(\mathbb{E}_p u(a, \cdot) + \mathbb{E}_p d)^2 + \frac{1}{2}.$$

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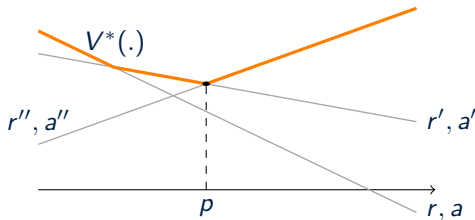
Necessary conditions

Value of information

Value of information function:

$$V^*(p) = \max_{a,r} \mathbb{E}_p V(r, a, \cdot)$$

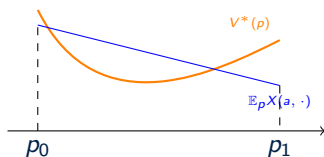
- is convex and
- it is strictly convex at p whenever there are multiple optimal actions



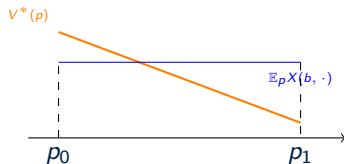
Necessary conditions

Value of information

- Value function $V^*(p) = \max_{a,r} \mathbb{E}_p V(a, r, \cdot)$ is convex.
- If a and b are best responses at beliefs p_0 and p_1 , then such configuration of expected reports imply that X cannot be incentivizable.



changing reports $\implies$ str. convex value



constant reports $\implies$ affine value

Necessary conditions

Example

Example

IQ test with = 2 True-False problems: $u(a, \theta) = 1\{a_1 = \theta_1\} + 1\{a_2 = \theta_2\}$.

What is the probability that both of your answers are incorrect?

$$X(a, \theta) = 1\{a_1 \neq \theta_1, a_2 \neq \theta_2\}.$$

	F_2	T_2
F_1	$0.5 - y$	y
T_1	$0.5 - x$	x

beliefs p_0

	F_2	T_2
F_1	$0.5 - z$	z
T_1	$0.5 - x$	x

beliefs p_1

$x, y, z < 0.25 \implies$ the subject is indifferent between (F_1, F_2) and (T_1, F_2)

$$E_{p_0} X(F_1, F_2) = x = E_{p_1} X(F_1, F_2)$$

$$E_{p_0} X(T_1, F_2) = y \neq z = E_{p_1} X(T_1, F_2)$$

Necessary conditions

Adjacency Lemma

If a and b are best responses at the same belief, and there is no other optimal action, we say that a, b are *adjacent*.

Adjacency Lemma

If X is incentivizable, then X is aligned with u on each pair of adjacent actions $\{a, b\}$:

For each adjacent pair a, b , there is $x \neq 0$ and y such that

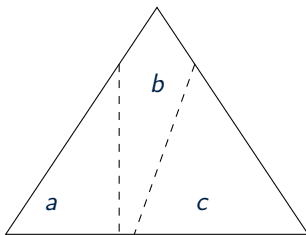
$$\bar{X}(a) = x\bar{X}(b) + y\Delta_a^b,$$

- where $\bar{X}(a, \theta) = X(a, \theta) - \frac{1}{|\Theta|} \sum_{\theta'} X(a, \theta')$, and
- $\Delta_a^b(\theta) = \bar{u}(b, \theta) - \bar{u}(a, \theta)$.

Representation

Adjacency graph

- Alignment on all actions is sufficient for incentivizability.
- Alignment on pairs of adjacent actions is necessary.
- To close the gap between sufficient and necessary conditions, we look at the adjacency graph.



belief simplex $\Delta\Theta$

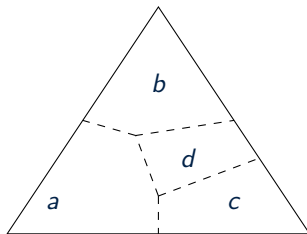


adjacency graph

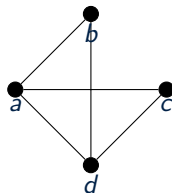
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adjacency graph

Necessary conditions

Adjacency paths

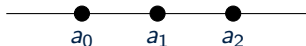


- For adjacent a_0, a_1 , there exist $x_1 \neq 0, y_1$ st.

$$\bar{X}(a_0) = x_1 \bar{X}(a_1) + y_1 \Delta_{a_0}^{a_1}$$

Necessary conditions

Adjacency paths



- These conditions carry over through adjacency paths ...

$$\begin{aligned}\bar{X}(a_0) &= x_1 \bar{X}(a_1) + y_1 \Delta_{a_0}^{a_1} \\ &= x_1 x_2 \bar{X}(a_2) + x_1 y_2 \Delta_{a_1}^{a_2} + y_1 \Delta_{a_0}^{a_1}\end{aligned}$$

Necessary conditions

Adjacency paths

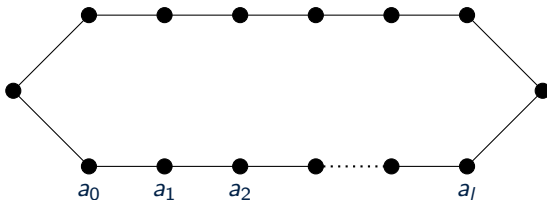


- These conditions carry over through adjacency paths ...

$$\bar{X}(a_0) = x_1 \dots x_l \bar{X}(a_l) + \sum_{0 < i \leq l} x_1 \dots x_{i-1} y_i \Delta_{a_{i-1}}^{a_i}$$

Necessary conditions

Adjacency cycles

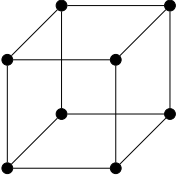
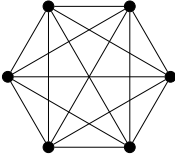


- ... and come back through cycles for $x = x_1 \dots x_n$:

$$\bar{X}(a_0) = x_1 \dots x_{n+1} \bar{X}(a_0) + \sum_{i=1, \dots, n+1} x_1 \dots x_{i-1} y_i \Delta_{a_{i-1}}^{a_i}$$

- The adjacency cycles provide more information if there are sufficient independence assumptions on Δ_a^b vectors.

Necessary conditions

	Tree	Product problem	Complete graph
Adjacency graph			
Examples	monotone problems, cognitive uncertainty (Enke and Graeber, 2023),	random problem selection (Azrieli et al., 2018), test with ≥ 3 questions	multiple choice question, prediction problems
Representation (necessary = sufficient)	aligned on each adjacency pair (Adjacency Lemma)	$\sum_i \gamma_i u_i(a_i; \theta_i) + d(\theta)$ and equivalents*	$u(a, \theta) + d(\theta)$, $d(\theta)$ and equivalents*

*Additional independence assumptions on payoffs

Special cases: Tree



Theorem: Incentivizability on tree-like problems

Suppose that the adjacency graph is a tree.

Then, X is incentivizable if and only if it satisfies the Adjacency Lemma for each adjacent pair.

- Proof: pasting together scoring rules over two disjoint set connected by a single adjacent pair.
- Example: monotone decision problems

Enke and Graeber (2023))

- DM chooses an estimate of some objective probability
 $A = \Theta = \{0, 1/n, \dots, 1\}$ incentivized with

$$u(a, \theta) = -(a - \theta)^2.$$

- What is the probability that the *ex post* correct CE is within ϵ of the chosen CE:

$$X(a, \theta) = \begin{cases} 1 & |a - \theta| < \epsilon \\ 0 & \text{otherwise.} \end{cases}$$

- a measure of cognitive confidence, not incentivized
- Adjacency Lemma $\Rightarrow X$ is not incentivizable for sufficiently large n .
- But, we could incentivize a question about expected payoffs/regret instead:

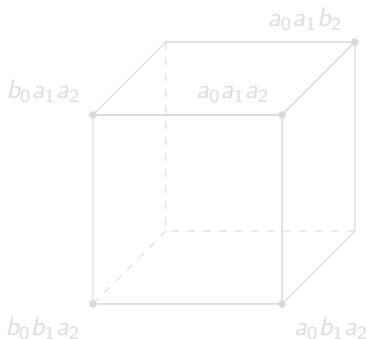
$$X'(a, \theta) = 1 - u(a, \theta).$$

Special cases: product problems

- $\Theta = \times_i \Theta_i$, $A = \times_i A_i$, where $(\Theta_i, A_i, u_i(.,.))$ is a collection of tasks, and

$$u(a, \theta) = \sum_i u_i(a_i, \theta_i),$$

- Example: Random problem selection, test with multiple problems
- Two actions $a, b \in A$ are adjacent if they differ in exactly one task: $a_{-i} = b_{-i}$ for some i

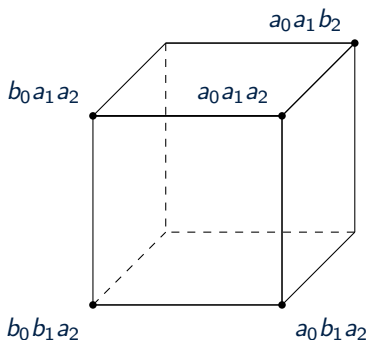


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Special cases: Product problems

Theorem: Incentivizability on product games

Suppose that

- 1 each task i is either binary ($|A_i| = 2$), or it has complete graph and vectors $\{\Delta_{a_i}^{b_i}, \Delta_{a_i}^{c_i}\}$ are linearly independent for all $a_i, b_i, c_i \in A_i$.
- 2 X depends non-trivially on at least 3 problems

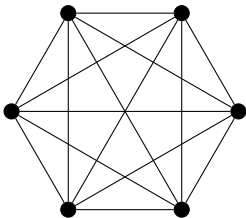
Then, X is incentivizable iff it is *product-aligned*: there are parameters $\gamma_i \in \mathbb{R}$ and $d \in \mathbb{R}^\Theta$ such that X is equivalent to for each a

$$\sum_i \gamma_i u_i(a_i, \theta_i) + d(\theta)$$

- Constants γ_i do not have to be the same, or have the same sign.
- For example "What is the expected difference between Macro and Micro part of the ECO101 exam?"

Special cases: Complete graph

- Decision problems with complete graphs have lots of cycles.
- Example: multiple-choice question.



Theorem: Incentivizability on complete graphs

Suppose that $|A| \geq 4$, the adjacency graph is a complete, and for all actions a, b_0, b_1, b_2 , vectors $\Delta_a^{b_0}, \Delta_a^{b_1}, \Delta_a^{b_2}$ are linearly independent.

Then, X is incentivizable if and only if it has aligned representation.

- What if the subject is allowed to ask more questions?

Joint incentivizability

Questions $X, Y : A \rightarrow \mathbb{R}^\Theta$ are *jointly incentivizable* if there exists $V : \mathbb{R}^2 \times A \times \Theta \rightarrow [0, 1]$ st. for every $p \in \Delta(\Theta)$,

$$\arg \max_{a, r, s} \mathbb{E}_p V(r, s, a, \theta) = \left\{ (a, \mathbb{E}_p X(a; \theta), \mathbb{E}_p Y(a; \theta)) : a \in \arg \max_{b \in A} \mathbb{E}_p u(b; \cdot) \right\}.$$

- A version of Adjacency Lemma holds.
- With $|\Theta| - 1$ questions, everything is incentivizable
- Chen et al. (2026) have a complete characterization for general $1 \leq k \leq (|\Theta| - 1)$ questions

Conclusions

Dynamic problems

Incentivizable questions X in dynamic problems (Pęski, Stewart (2026)):

	Action-independent information	Actions may affect the quality of information	Actions may affect the states
$t = 1$	$\gamma u^*(a_1, p) + f(p)$ PS25 conditions	$\gamma u^*(a_1, p) + l(p)$ nontrivial continuation	$\gamma u^*(a_1, p)$ always
$t = 2$	$\gamma u_2(a_2, \theta) + d(\theta)$ only if payoffs are <u>separable</u>	$\times$	$\times$

- $\gamma = 0, 1$
- $f(p)$ is arbitrary (sufficiently regular) function of beliefs
- $l(p)$ is affine in p

- Sufficient conditions: Aligned questions (i.e., questions about affine transformations of payoffs) are incentivizable.
- Necessary conditions: Adjacency Lemma.
- **"Theorem"** In three classes of decision problems, question X is incentivizable if and only if it satisfies the Adjacency Lemma.
- Special representations when the adjacency graph is complete, it's a tree, or in product problems.
- Other questions:
 - dynamic elicitation, learning,
 - action-dependent states ("moral hazard"),
 - "robust" elicitation.